a

Task

Colour fundus photograph in, four lesion probability maps out

- **MA** microaneurysm
- **HE** hemorrhage
- **EX** hard exudate
- **SE** soft exudate

Data collection

7 test sources · 5,121 images

Training sources

IDRiD	81
DDR	757

Independent sources

TJDR-std	257
TJDR-uwf	304
MAPLES-DR	198
FGADR	1,842
Retinal-Lesions	1,593
DiaRetDB1	89

Model collection

39 variants in the registry

- Tier 1 · main table — **22**
- Tier 2 · if time allows — **6**
- Tier 3 · paper audit only — **11**

Reproducibility level R0–R4 recorded for every variant; failures stay in the denominator

Four evaluations

- **Paper audit** *L1* — numbers as reported, and where they came from
- **Author weights** *E1* — released checkpoints, three numbers each
- **Author recipe** *E1r* — retrained with the authors' own pipeline
- **Unified retraining** *E2* — shared data, budget, augmentation, selection

Tests and metrics

Internal test — official test set of the training source

Independent test — sources never used for training

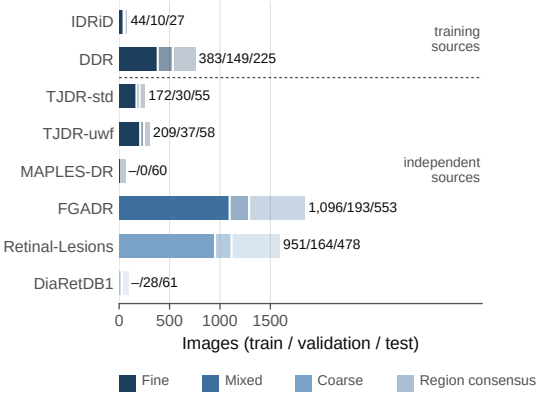
Primary — exact pooled AUPR inside the field of view; mAUPR

Secondary — Dice, IoU; lesion-level recall by lesion size

Statistics — 3 seeds; image bootstrap; Friedman + Nemenyi

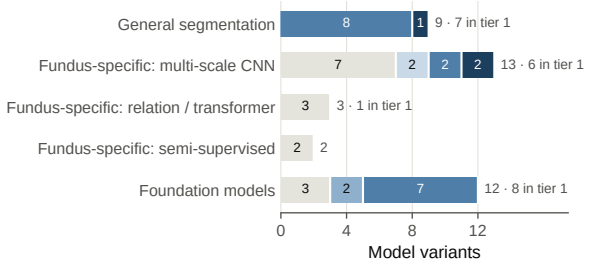
b

Test sources and frozen splits



c

Registered models by family and reproducibility level

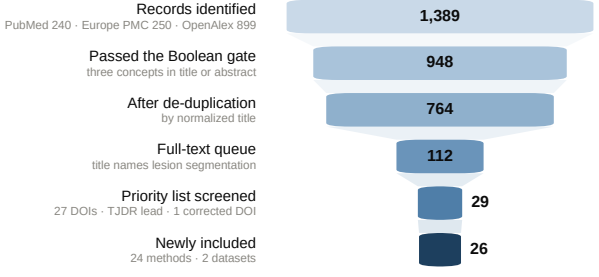


Reproducibility level

- R0 no code
- R1 model definition
- R2 inference + weights
- R3 training code
- R4 code + weights + splits + env.

d

Literature search and screening



Fixed search 2015-01-01 to 2026-09-30; with the snowballed corpus the ledger holds 86 records.

e

Factors assessed

Training source *E3b*	Label granularity *E4*	Training amount *E8*	Input resolution *E7*
IDRiD · DDR · TJDR-std · FGADR	fine · mixed · coarse, alone and combined	10–100% of DDR, merged sources	512 · 1,024 · 1,440 · native patches

Loss *E9*	Extra input *E6*	Foundation models *E5*	Training recipe *E2b, E2c*
CE · Dice · CE + Dice · bin loss	image alone or with vessel map	zero-shot · oracle prompt · PEFT · encoder + decoder	source recipe (B1); learning rate ×⅓, 1, 3

One factor changes at a time; everything else stays fixed. E3b–E9 use five representative models: U-Net, nnU-Net, M2MRF, SAM2-UNet, RETFound + UPerNet.